

CLIMATE RESILIENCE IN HIMALAYAN TOURISM

A DATA-DRIVEN CASE STUDY OF RAINFALL, LANDSLIDES & TOURISM IN UTTARAKHAND

ABSTRACT

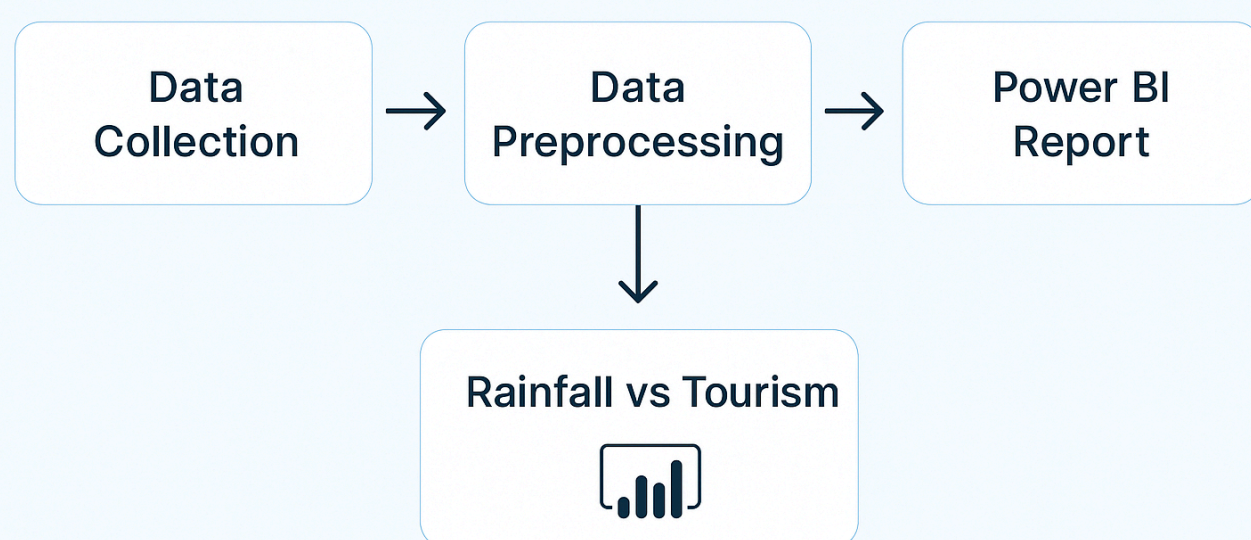
- To discuss the effects of climate on the Himalayan tourism based on actual data.
- To determine the relationship between landslides, rainfall, and the inflow of tourists.
- To gauge the challenges to the resilience of the tourism industry in Uttarakhand.

OBJECTIVE

- To examine the annual tourism patterns in Uttarakhand.
- To analyze the climate trends using the rainfall and landslide data.
- To determine the high-risk tourism areas under the influence of the variability of the climate.
- To create Power BI figures on climate tourism.

DATA PROCESSING STEPS

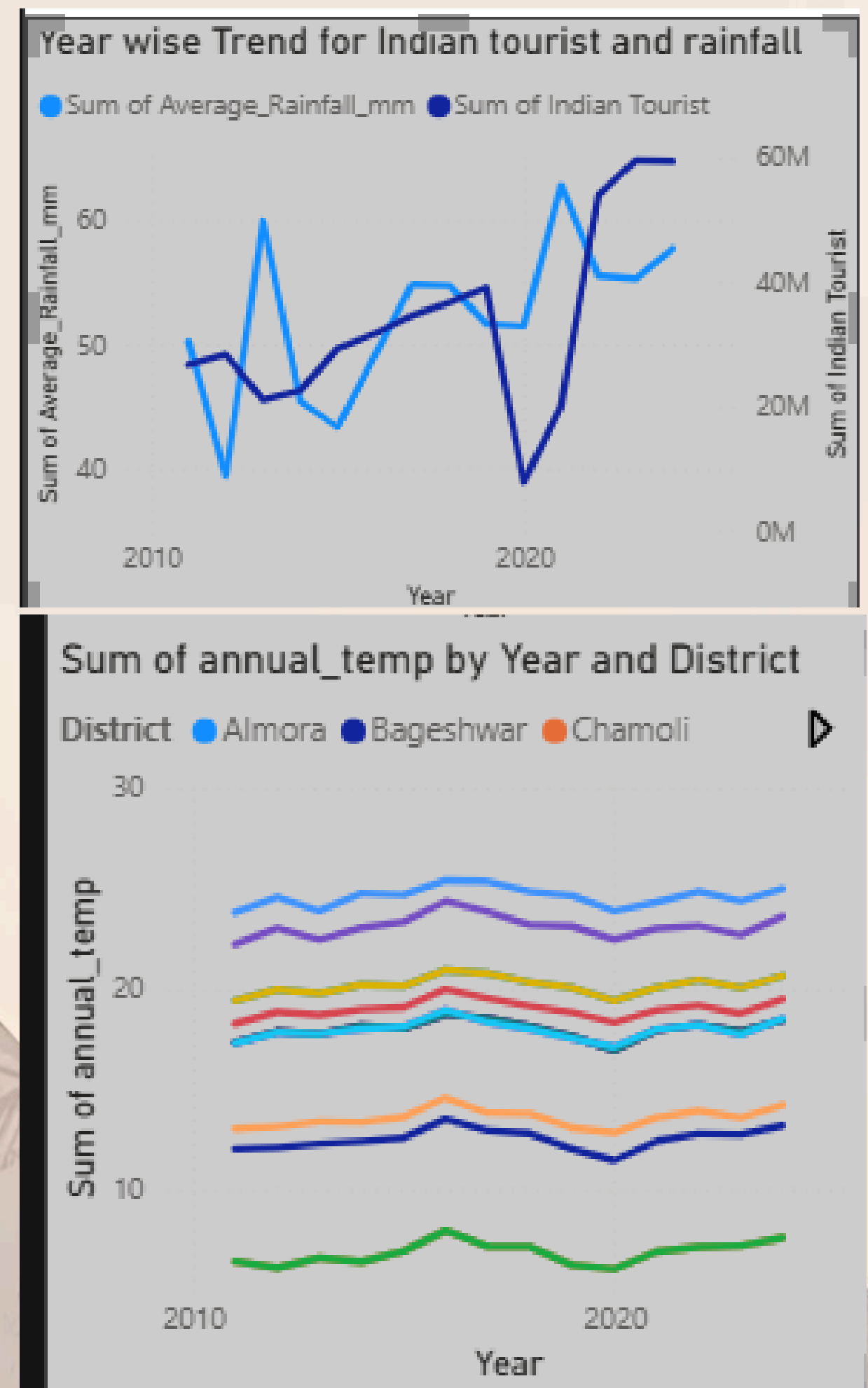
Data Processing Steps



Gathered tourism, rainfall, and landslide data
→ cleaned and standardized values →
unpivoted tourism years → constructed
Year-wise model → created power BI
dashboard to analyze.

Sources: Government of India – IMD, NRSC-ISRO,
Uttarakhand Tourism ,NASA

DASHBOARD



MOTIVATION AND CONCLUSION

- Climate hazards repeatedly disrupt tourism in Himalayan states, with major disasters in Uttarakhand (2013, 2021, 2022) highlighting serious safety risks.
- The study supports SDG 11 & SDG 13 by using evidence-based analysis to identify vulnerabilities and promote climate-resilient, safer tourism planning.
- Climate hazards significantly influence tourism in Uttarakhand, with rainfall extremes and landslides causing major disruptions in key years. The analysis highlights vulnerable districts and shows how tourism trends are closely tied to climate patterns.

Strengthening climate resilience—through data-driven planning and sustainable practices—is essential for ensuring safe and stable Himalayan tourism in the future.

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